1 Integrals of Exponential Functions

1. Use the derivative formulas $\frac{d}{dx}e^x = e^x$ and $\frac{d}{dx}a^x = a^x \ln(a)$ to find antiderivative formulas for exponential functions. Check your answers by differentiating.

   (a) $\int e^x dx =$

   (b) $\int a^x dx =$

2. Evaluate $\int e^{-x} dx$.

3. Evaluate $\int x^2 e^{-2x^3} dx$.

4. Evaluate $\int e^x \sqrt{1+e^x} dx$.

5. Evaluate $\int e^x(3e^x-2)^2dx$.

6. Evaluate $\int 3x^2e^{2x^3}dx$.

7. Evaluate $\int_1^2 e^{1-x}dx$.

8. Evaluate $\int_0^2 e^{2x}dx$.

9. Evaluate $\int_1^2 \frac{e^{1/x}}{x^2}dx$.

10. Evaluate $\int_1^2 \frac{e^{4x^{-2}}}{x^3}dx$.

2 Integral of $1/x$

Consider $\int x^\alpha dx$, $\alpha \in \mathbb{R}$. The derivative rule $\frac{d}{dx}x^\alpha = \alpha x^{\alpha-1}$ implies that

$$\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1}, \quad \alpha \neq -1.$$

What about $\alpha = -1$? That is, what is $\displaystyle\int \frac{dx}{x}$?

11. Use the derivative formula $\frac{d}{dx}\ln|x| = \frac{1}{x}$ to evaluate $\int \frac{dx}{x}$. Check your answer by differentiating.

12. Evaluate $\displaystyle\int \frac{1}{x+2}dx$.

13. Evaluate $\int \frac{3}{x-10}dx$.

14. Evaluate $\int \frac{2x^3+3x}{x^4+3x^2}dx$.